

A Quick Forray into Concurrency with Pythons AsyncIO

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Independent

Microservice - A microservice is a small, independently deployable software component that handles one specific business function within a larger application. Instead of building a monolithic app that does everything, you break the system into many small services - each with its own codebase, database, and deployment lifecycle. They communicate with each other over a network, typically via HTTP/REST .

gRPC - A high-performance remote procedure call framework developed by Google. The core idea is that calling a function on another service should feel as close as possible to calling a local function. Uses protobuf (protocol buffers, think like JSON or XML) Protobuf is binary, gRPC runs over HTTP/2 which supports multiplexing multiple calls over one connection. More ceremony, REST with JSON is easier to debug, but when performance matters, gRPC is ideal.

Microservices as an Idea

I am skeptical of microservices due to the large integration cost of combining these many different parts together. If I were to make a guess, Bosch was following this model on DAT3 while I was working there. All I heard was the integration complexity and the endless versioning problems.

The case for microservices

When the teams are very large (i.e >200 contributors) it naturally makes sense to break them off into smaller chunks at times. According to Claude, it is likely Apple utilizes a hybrid system, where there are a number of large monoliths that must interact with each other. I picture this as a number of tall buildings in a city all whom are interacting with each other, whom all may possess totally unique constructions, but some buildings must make connections so they build skyways, and share electricity. Contrast this to the microservice model in my head, microservices live more like planned housing, there are some different types of models, but they all feel samey, and likely are built into one of a number of molds. I'm not sure if this is a totally accurate representation, but at this moment this is my model of the world (*The really cool thing is, if I learn new information I can change this model pretty easily!*)

Crux: AsyncIO

Understand the difference between CPU bound work and I/O bound work, what the Python Global Interpreter Lock is (GIL), and how to utilize non-blocking I/O with an event loop.